

Sparse DANP for FPGA-Oriented Online Learning

Internal Progress Report

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This report summarises completed experiments on sparse node perturbation under the Decorrelated Activity-based Node Perturbation (DANP) framework. All results are measured values read directly from the `results/` directory. Architecture used throughout (unless otherwise stated): 3-hidden-layer MLP with `hidden_sizes = [1024, 1024, 1024]`, trained on MNIST or CIFAR-10 with Adam, cross-entropy loss, 3 random seeds.

1 Baseline Sparse Experiments

The total node count for the 1024-wide architecture is $3 \times 1024 + 10 = 3082$. At 25% sparsity this gives $3 \times 256 + 2 = 770$ active nodes; at 50% sparsity, $3 \times 512 + 5 = 1541$. The column *Rel. Cost* reports the fraction of nodes that receive noise and a weight update relative to full perturbation. A dash (—) indicates the experiment was not run.

1.1 MNIST

Table 1: MNIST results. Architecture: `[1024,1024,1024]`, 3 seeds. The column *Ep.* gives the number of training epochs.

Policy	Fraction	Active/Total	Rel. Cost	Final Test Acc.	Ep.
Full DANP	—	3082/3082	1.000	0.9642	20
Random	50%	1541/3082	0.500	0.9407	5
Random	25%	770/3082	0.250	0.9558	20
Scheduled	50%	—	—	—	—
Scheduled	25%	770/3082	0.250	0.9566	20
ActivityDiff TopK	50%	1541/3082	0.500	0.9377	5
ActivityDiff TopK	25%	770/3082	0.250	0.9313	5
ActivityLoss TopK	50%	1541/3082	0.500	0.9398	5
ActivityLoss TopK	25%	770/3082	0.250	0.9314	5
GradientAligned TopK	—	—	—	—	—

1.2 CIFAR-10

2 Gradient Distribution Analysis

The `--analyze_gradient_dist` flag was used to measure how concentrated the per-node DANP effective-error signal $g_i \propto \Delta a_i \cdot \Delta L / \|\Delta \mathbf{a}\|^2$ is across all hidden and output nodes. Statistics are

Table 2: CIFAR-10 results. Architecture: [1024,1024,1024], 5 epochs, 3 seeds.

Policy	Fraction	Active/Total	Rel. Cost	Final Test Acc.
Full DANP	—	3082/3082	1.000	0.4094
Random	50%	1541/3082	0.500	0.4080
Random	25%	770/3082	0.250	0.4070
Scheduled	50%	1541/3082	0.500	0.4122
Scheduled	25%	770/3082	0.250	0.4102
ActivityDiff TopK	—	—	—	—
ActivityLoss TopK	50%	1541/3082	0.500	0.4109
ActivityLoss TopK	25%	770/3082	0.250	0.4055
GradientAligned TopK	50%	1541/3082	0.500	0.4117
GradientAligned TopK	25%	770/3082	0.250	0.4076

means over 3 seeds. **CV** is the coefficient of variation of $|g_i|$; **Top 10%** is the fraction of total $|g|$ energy carried by the top-10% of nodes; **Gini** is the Gini coefficient of $|g_i|$. All three statistics are near zero throughout training on both datasets.

2.1 MNIST

Table 3: Gradient distribution statistics on MNIST (mean across 3 seeds). Architecture: [1024,1024,1024], 5 epochs, full DANP.

Epoch	CV	Top 10%	Gini
1	0.238	0.135	0.134
2	0.253	0.138	0.142
3	0.263	0.141	0.148
4	0.272	0.143	0.152
5	0.281	0.144	0.156

2.2 CIFAR-10

Table 4: Gradient distribution statistics on CIFAR-10 (mean across 3 seeds). Architecture: [1024,1024,1024], 5 epochs, full DANP.

Epoch	CV	Top 10%	Gini
1	0.228	0.132	0.129
2	0.234	0.133	0.132
3	0.238	0.134	0.134
4	0.241	0.134	0.136
5	0.245	0.135	0.138

The gradient signal appears relatively diffuse (low CV, low Gini, low Top 10% energy concentration), which may explain why adaptive node-selection policies do not significantly outperform random sparse perturbation.

3 Width Sweep

To assess whether the accuracy–sparsity tradeoff generalises across network sizes, the same experiment was repeated on MNIST at three hidden-layer widths. Each run uses 20 epochs, 3 seeds, and the Scheduled 25% policy as the sparse baseline. The full DANP reference is run with the same epoch count and seed set.

Table 5: Width sweep on MNIST (Scheduled 25%, 20 epochs, 3 seeds). *Full Acc.* and *Sparse Acc.* are final test accuracies.

Architecture	Total Nodes	Active/Total	Rel. Cost	Full Acc.	Sparse Acc.	Acc. Drop
[64, 64, 64]	202	50/202	0.248	0.9391	0.9342	−0.0049
[256, 256, 256]	778	194/778	0.249	0.9530	0.9480	−0.0050
[1024, 1024, 1024]	3082	770/3082	0.250	0.9642	0.9566	−0.0076

4 Conclusions

- Sparse DANP can substantially reduce perturbation and update cost while preserving most of the full DANP accuracy.
- With only 25% active nodes, accuracy degradation remains below 1% across the tested MNIST network widths.
- Simple random and scheduled sparse policies perform comparably to more complex adaptive selection methods.
- Adaptive top-k policies based on activity difference, activity-loss contribution, or gradient-aligned scores did not show a consistent advantage.
- Gradient-distribution measurements suggest that the DANP learning signal is broadly distributed across nodes, which may explain why simple sparse policies are effective.
- These results indicate that FPGA-friendly sparse perturbation may not require expensive ranking or sorting logic.